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Digital transformation of business ecosystems: Evidence from the Korean pop industry

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Abstract

The notion of business ecosystems is gaining prominence among academics and practitioners. Scholars advise that business ecosystems are inherently fluid, unbounded and amorphous and thus that their boundaries can shift. Practitioners further suggest that business ecosystems are characterised by inter-network – as opposed to inter-firm – competition, and, moreover, that they are mainly driven by technological advancements. And yet few studies examine the role of information technology (IT) in driving boundary practices that enable the formation and transformation of business ecosystems. Through an in-depth case study of technology-enabled transformations within the Korean pop music (K-pop) industry, our study examines how the digital transformation of business ecosystems unfolded. Our study contributes to the emergent body of knowledge on business ecosystems in a number of ways. Our investigations uncover the conditions under which the constituent firms operate and analyse the role of IT and its implications on new organisational forms. From our analysis, we present a framework that reveals insights on critical boundary practices and formative strategies for digital business ecosystems. We illustrate how these boundary practices drive industry change, from a largely linear content delivery system resembling a value chain to a new value network of co-specialisation and self-organisation among firms in a new digital business ecosystem.

KEYWORDS

boundary spanning practice, case study, digital business ecosystem, Korean pop

1 | INTRODUCTION

Business ecosystems represent a network of entities with differing interests bound together as a collective whole, such that the fate of its members is bound to the structure of that network and the roles played by its members (Iansiti & Levien, 2004a). These entities share connections among communities, organisations, processes and technologies across industries (Moore, 1996). In doing so, business ecosystems often serve as the foundation for new business strategies and inter-organisational collaboration between their constituent firms, and thus redefine how businesses approach competition and innovation today (Lyytinen, Yoo, & Boland, 2016). According to practitioners, business ecosystems consist of communities that create new value through sophisticated models of collaboration and competition (Kelly, 2015). This reality reflects how businesses are no longer driven solely by economies of scale but instead also by the 'economics of networks' (Shapiro & Varian, 1999). In other words, it is not individual firms, but the networks that firms lead that are engaged in business competition with one another in the present networked economy (Tan, Pan, Lu, & Huang, 2015). Consequently, the development and management of business ecosystems is a topic that has attracted increasing attention from researchers and practitioners over the last few years (Jacobides, Cennamo, & Gawer, 2018; Wang, Lee, Meng, & Butler, 2016).

Organisational boundaries are central as they pose significant issues around employment and participation in business ecosystems. Despite the growing number of studies, few studies exist on organisational boundary issues in business ecosystems (Santos & Eisenhardt, 2005). The transformation of business ecosystems is inevitable due to their fluid, unbounded and amorphous nature (Iansiti & Levien, 2004a). Yet if not adequately managed, transformation can lead to envelopment (Eisenmann, Parker, & Van Alstyne, 2011), which may threaten the viability of the entire ecosystem. Furthermore, little attention has been paid to the exact role of information technology (IT) in the formation of healthy business ecosystems, with the exception of Kim, Lee, and Han's (2010) guidelines for determining the health, strategies and roles of IT in business ecosystems. This situation has led scholars to call for increased efforts to address the processes and implications of business ecosystem formation, which may comprise new forms of digital moves, processes, collaboration and infrastructures (Bharadwaj, El Sawy, Pavlou, & Venkatraman, 2013; Markus & Loebbecke, 2013).

To bridge the above knowledge gaps, we ask: '*how does IT enable effective boundary practices in the transformation of a business ecosystem?*' To address this question, we conducted a case study on South Korean pop music (K-pop) and examined the digital transformation of this business ecosystem. The K-pop ecosystem comprises performing artists, entertainment companies, the media, digital platforms and event management firms. This business ecosystem generates over 40 billion USD and attracts over 30 million fans worldwide. South Korean music and self-styled acts (eg, PSY, BTS, Blackpink) and entertainment platforms (eg, SM Entertainment) have recently garnered widespread global attention in bringing their style to the American mainstream. According to *The New York Times* in May 2018, for the first time in a decade, BTS became the first K-pop act to reach the top of America's album Billboards charts.[1]

Our empirical investigation focuses on the digital strategies adopted by diverse participants to deliver creative content to global consumers. We further investigate how these strategies impact the transformation of the business ecosystem. To do so, we examine the implications of new IT, processes and boundary practices (BP) that enable creative businesses to develop. Concomitantly, we adopt the body of knowledge on boundary spanning as a guiding theory (Klein & Myers, 1999) to examine the transformation of businesses and technologies in this business ecosystem.

Our study provides several empirical contributions (Ågerfalk, 2014) by addressing the lack of real case studies on the digital transformation of business ecosystems. Given that there have been little to no a priori conceptualisations of this topic, our study also advances theory on how technology enables the development of creative content and its business ecosystem. Our study has profound implications for understanding business ecosystem transformations and the scaling of vertical markets across different IT infrastructures and platforms for the production of creative content. Moreover, our in-depth case study serves to identify the social, economic and technological processes that lead to the development of business ecosystems.

Building on Levina and Vaast (2006), this article unpacks the process by which IT shapes boundary spanning practices. In examining practices in a unique music and entertainment ecosystem, we contribute to the literature by showing how BP, actors and objects can go beyond their expected bridging role. For practice, the case study illustrates how companies can adopt and implement a business ecosystem strategy in an increasingly digital and mobile era (McKelvey, 2016; Tapscott, 2014) to overcome the threat of commoditisation and a host of issues around compatibility and computability. Moreover, our analysis extends beyond the value chain to reveal how businesses can leverage IT to coordinate resources found in overlapping entities to shape boundary spanning capabilities to form a value network and to develop a business ecosystem effectively.

The rest of the article is organised as follows. We first provide a background on business ecosystems. Then, we discuss the relevance of using BP, objects and technology to study business ecosystems. The next section presents the research method and data collection. We then provide a case description before discussing the case findings. Finally, we discuss the implications of our study for practice and theory before concluding the manuscript. Notably, our findings offer a starting point for the theoretical development of a processual and situated understanding of the conditions of practice-based IT instantiations in business ecosystems and their outcomes.

2 | BACKGROUND

This literature review summarizes and synthesizes studies of business ecosystems. In mapping the extant literature, we reveal its gaps and position our work in relation to them. We also discuss the position and guiding lens of our empirical study on digital transformation in business ecosystems.

2.1 | Business ecosystem organisation

In recent years, scholars have studied the notion of business ecosystems in different forms,[2] adopting various levels of analyses (Tsujimoto, Kajikawa, Tomita, & Matsumoto, 2017). A number of observations can be made from these studies.

First, as we note in our opening remarks, the literature maintains that business ecosystems characterise economic communities of interacting organisations and individuals (Moore, 1996). At minimum, business ecosystems include customers, suppliers, lead producers and other actors who interact with one another to produce goods and services (Moore, 1996). More recently, scholars have also highlighted the roles of platform-owners, complementors and intermediaries (Benlian, Hilkert, & Hess, 2015; Foerderer, Kude, Mithas, & Heinzl, 2018; Huang, Tafti, & Mithas, 2018). Such recent studies present the distinctive properties that characterise the interactions between interdependent actors in their creation and commercialisation of innovation (Jacobides et al., 2018).

Second, there are diverging views about the type of expected outcomes of a business ecosystem. Some ecosystems focus on value creation and capture, while other ecosystems support the development of an innovation or a specific value proposition. The latter perspective is commented on by Adner (2017), who explains that the 'ecosystem-as-structure' consists of an alignment of interacting actors and their activities, which are required for a value proposition to materialise. He contrasts his view with the common view of ecosystems found in IS research. Adner (2017) calls this view 'ecosystem-as-affiliation', a traditional view that is shared by numerous scholars (Iansiti & Levien, 2004a; Moore, 1996; Tiwana, 2013). A commonly held assumption is that a keystone, core firm or platform leader is present at the centre of a business ecosystem (Tan et al., 2015). The central point of the business ecosystem is considered an anchor to which all other stakeholders are attached. The keystone not only dominates and tends to capture the greatest value from the ecosystem but also directly influences the survival of the whole ecosystem and its members (Iansiti & Levien, 2004b). Along these lines, work on business ecosystems poses questions about how the properties of self-organisation, self-management and scalability materialize in this new form of technology-enabled economic coordination (Moore, 2006; Stanley & Briscoe, 2010) – questions inspired by analyses of natural ecosystems. We summarise the two types of business ecosystems in Table 1.

Third, self-organisation occurs among actors in a system when tensions are imposed on them and when they are motivated to change (Coleman & Henry, 1999; McKelvey, 2016). Self-organisation is a term typically used in the literature to describe a process in which small units assemble into larger structures without external intervention (Anderson, 1999; Peltoniemi & Vuori, 2004). Such processes come about when the units contain relevant information about the structure to be built within themselves and the right global conditions prevail. Understanding these processes enables the synthesis of complex organisations and processes (Mitleton-Kelly, 2003). Here, it is helpful to note that large and complex structures can be naturally *formed* from many small units in a bottom-up process. Self-organised business ecosystems possess the capabilities of novelty and transient networks, a heterogeneous structure, multiple network roles, interconnectedness and shared fate and the ability to absorb external shocks (Iansiti & Levien, 2004a). Correspondingly, business ecosystems are meta-organisational structures that manage the tension between the homogeneity of human actors and the heterogeneity of organisational resources, such that they surround, permeate and reshape markets and hierarchies (Moore, 2006).

Fourth, we observe from these studies that IT are internal and/or external stimuli for business ecosystems (Cui, Ouyang, Chen, & Li, 2019; Martínez de Aragón, Alonso-Zarate, & Laya, 2018). These can include internet-based technologies (eg, cloud computing, mobile technology) and networks (Katsamakas, 2014; Nachira, Dini, & Nicolai, 2007). In fact, digital business ecosystems are a specific type of business ecosystem supported by an IT infrastructure which facilitates cooperation and knowledge-sharing (Corallo, Passiante, & Prencipe, 2007; Nachira, Dini, et al., 2007; Tan et al., 2015). While it can be useful to identify IT that develop ecosystems (eg, as in Curry and Sheth (2018)), researchers are urged to pay more attention to the emergence and evolution of such ecosystems (Tiwana et al., 2010; Tsujimoto et al., 2017) and their transformation of industries (de Reuver, Sørensen, & Basole, 2018). Correspondingly, Wang (2018) urges scholars of business ecosystems to pay more attention to ecological and environmental factors.

While the above assertions are useful, there remains a deficit of theoretical examinations into the digital transformation of business ecosystems. At the outset, digital transformation characterises the complex changes that digital technologies can bring to business models, such as those that modify products, organizational structures or

TABLE 1 Organisation of business ecosystems

Nature of organisation	Description	References
Command and control	A keystone or focal firm has a more regulatory than cooperative presence. The more centrality a firm has, the more power it has.	Bonabeau and Meyer (2001)
Self-organised	Firms in the ecosystem possess a more open approach to cooperation. Multi-stakeholder partnerships are often formed through incentives for work projects.	Williamson and De Meyer (2012)

processes (Hess, Matt, Benlian, & Wiesböck, 2016). To be sure, researchers highlight the need to expand existing knowledge on how contemporary business ecosystems cultivate the alignment of digital transformation strategies, infrastructure, governance and business value (Hess et al., 2016; Tiwana, 2013). One way is to consider the networked, multi-level, boundary-spanning characteristics of ecosystems (Foerderer et al., 2018; Huber, Kude, & Dibbern, 2017; Tiwana et al., 2010). This can include examining how individual firms transform (Valkokari, 2015) and how firms develop an interdependent network (Lee & Alnahedh, 2016) during digital transformation. In fact, scholars such as Adner (2017) urge researchers to pay attention to the alignment of activities surrounding a focal firm's value proposition, situating an ecosystem as '*an aligned structure of a multilateral set of partners that need to interact in order for a focal value proposition to materialize*' (Adner, 2017; Hannah & Eisenhardt, 2018). Such an undertaking is especially significant when an organisation emerges as a complex social system that consists of interdependent subsystems. Special arrangements will be required to manage the blending of structural configuration logics and different operational principles in organisations (Battilana & Dorado, 2010; Kortmann, 2012) during the accumulation of external resources for collaboration and integration across and within boundaries (Billis, 2010; Ménard, 2004).

In summary, our literature review reveals that the archive lacks theoretical scrutiny and that little work has been done on the digital transformation of business ecosystems. Research that addresses these gaps is likely to have significant implications for focal firms that provide a set of tools and services allowing various complementors to join and add value to the focal value proposed by the firms in business ecosystems (Bergman, Lyytinen, & Mark, 2007; Krishnan, Rai, & Zmud, 2007; Stanescu et al., 2010). Moreover, such a study would reveal valuable lessons on interdependent networks (Lee & Alnahedh, 2016), complementarity (Alaimo, Kallinikos, & Valderrama, 2019; Breznitz, Forman, & Wen, 2018) and tensions between control and generativity (Lusch & Nambisan, 2015), which can inhibit synergistic relationships within business ecosystems. In addition, the digital transformation of business ecosystems presents an opportunity for researchers to explore notions relating to boundary objects (BO) and practices to foster effective collaboration, innovation and/or competitive advantage (Bergman et al., 2007; Krishnan et al., 2007; Stanescu et al., 2010). Reiterating our research agenda, we refer to BP as our guiding lens for our in-depth examination of the digital transformation of a business ecosystem, on which we elaborate in the next section.

2.2 | BP and technology

Boundary spanning, which generally describes, but is not limited to, the transfer of information from the outside of an organisation to the inside (Marrone, Tesluk, & Carson, 2007; Tushman & Scanlan, 1981), engenders rich discussions across the areas of management, organisational science and IS. To date, studies have reported the attributes and abilities of boundary spanning individuals (eg, Dollinger, 1984) and the role of leadership in creating the boundary spanning effect in communities (eg, Fleming & Waguespack, 2007) and IT-enabled boundary resources that stimulate value creation and reconfigure boundary relations (Barrett, Davidson, Prabhu, & Vargo, 2015; Levina & Vaast, 2005).

Studies also have introduced boundary spanning mechanisms, including increasing the competence of personnel (Bassellier, Benbasat, & Reich, 2003) and creating integrative boundary spanning roles (Levina & Vaast, 2005; Tushman & Scanlan, 1981). The literature refers to the mechanism of boundary spanning, in which IT does not transform practices by itself but is instead part of a complex series of changes in practice that include the roles of intermediaries, objects and actors (Levina & Vaast, 2006).

In examining BP, two notions are central. First, BO refer to artefacts used by actors to acquire both local usefulness (which actors jointly recognise and value) and a common identity in practice. In this research, we posit that BOs and digital infrastructures are similar concepts. Second, *boundary spanners (BS)* refer to organisational actors who establish the local usefulness of BO-in-use as well as their common identity.

To better understand the emergence of organisational competence in boundary spanning, we need to investigate how actors in the ecosystem become BS in practice by drawing on their nomination or by looking at them independent of their expected roles (Levina & Vaast, 2005, 2006). A practice-based view has emerged from the strategic management field, enabling researchers to study how the firm implements organisational practices in a given context and to draw a better understanding of the business value of IT (Peppard, Galliers, & Thorogood, 2014; Whittington, 2014). Practice, 'a defined activity or a set of activities that a variety of firms might execute' (Bromiley & Rau, 2014, p. 1249), is a central part of this view.

Focusing on the mode of practice production, we can better understand 'how IT is incorporated into organizational practices in different ways so as to either reinforce or help bridge existing boundaries within and across organizations' (Levina & Vaast, 2006, p. 47). Levina and Vaast (2006) argue that any practice involves varying degrees of embodiment (ie, relying on personal relationships) and objectification (ie, relying on the exchange of objects). Along these lines, IT is a medium for sharing objects in the production of practices, such that IT either reinforces institutional boundaries or helps bridge them.

The development of business ecosystems poses many strategic questions about cross-industry convergence and boundaries, such that researchers are urged to explore a deeper understanding of organisational boundary issues that exist in business ecosystems (Santos & Eisenhardt, 2005). According to Santos and Eisenhardt (2005), researchers should emphasise non-efficiency perspectives and relationships and more problem-based research.

Tilson, Lyytinen, and Sørensen (2010) posit that a careful examination of processes and IT infrastructures that overlap, reinforce or compete across business ecosystems reveals how digitisation and the actualisation of actors and artefacts can reshape an organisation and its services. These infrastructures represent new IT artefacts, that is 'the basic information technologies and organizational structures, along with the related services and facilities necessary for an enterprise or industry to function' (Tilson et al., 2010, p. 1).

According to Whyte and Lobo (2010, pp. 557, 565), IT links to the rich work on BO, such that the 'reconceptualization of BO as a digital infrastructure' provides an opportunity to enrich the discussion of BP used in coordination across boundaries. Hence, a larger unit of analysis can be used to examine processes and artefacts that span vertical boundaries – including technical specifications designed to promote coordination within or across industry sectors so that no break in the communication paradigm occurs (Markus, Steinfield, & Wigand, 2006; Steinfield, Markus, & Wigand, 2005) – and horizontal boundaries to coordinate and integrate businesses, agencies and markets (Franke Kleist, Hanson Frieze, & King, 1999; Layne & Lee, 2001).

3 | RESEARCH METHOD

We adopted an interpretive case study research approach (Klein & Myers, 1999; Walsham, 1995) to address our research question and to examine how IT drives industry transformation and enables the development of business ecosystems. The case study research approach is appropriate due to the exploratory nature of our study (Siggelkow, 2007) and allows us to examine industry processes in an in-depth manner and address the 'how' of our research question (Walsham, 1995).

Furthermore, the phenomena of emergent business ecosystems are complex, multi-faceted (Koch & Schultze, 2011) and embedded within an industry context that makes examining them through stakeholders' interpretations more suitable than using a quantitative approach (Klein & Myers, 1999). Our interpretivist case study approach seeks to provide empirical contributions (Ågerfalk, 2014) and reveal insights into a novel phenomenon without relying explicitly on a priori conceptualisations (Mays & Pope, 2000). We found no established theoretical model available that accounts for complex relations between IT and the formation of business ecosystems to guide our study a priori (Klein & Myers, 1999). Building on our existing knowledge of business ecosystems, we contacted multiple stakeholders and actors in a business ecosystem, conducted fieldwork at multiple case study sites, made case observations using a boundary spanning theoretical lens and performed a qualitative data analysis. Our approach allowed for new, unexpected and in-depth findings unknown at the outset of the inquiry to emerge from the data (Pan & Tan, 2011).

We used several criteria to select our case study. First, we established that the case needed to allow us to examine core firms that lead and organise their business ecosystem and the technologies that support interoperable business processes across ecosystems. This allowed us to better understand the conditions for BP. Second, the case should reflect a competitive environment that necessitated digital transformations and new organisational forms. This allowed us to foster a better understanding of the strategic nature of overlapping ecosystems. Third, the actors studied needed to be part of, and possibly orchestrate, business ecosystems, so we could study the underlying boundary spanning mechanisms that should be applied in this context. Last, the case organisation needed to have internal structures and external operations sophisticated enough to enable the study of the practices and capabilities (both IT and non-IT enabled).

For the above reasons, we chose to examine the K-pop industry. K-pop is part of *Hallyu* ('the Korean wave'), which is today a large business ecosystem globally renowned for its creativity and innovation in creative media and entertainment content. In Korea, the media and entertainment industries represent an intersection between Arts and Culture and Business and Technology. Previously, the K-pop industry sought to address poor customer experiences and returns from the industry due to boundaries created by the numerous vertical markets and standards determined by incumbent video platform providers, video service providers and video application providers as well as by horizontal markets (eg, broadcasters, media and entertainment). The industry achieved phenomenal success in this regard through the IT-driven development of diverse business ecosystems, making the industry a revelatory case (Gerring, 2009) for our study.

3.1 | Case background

Hallyu, literally translated as 'the Korean Wave', was a term coined in 1999 by China's *Beijing Youth Daily* to describe the growing popularity of South Korean pop culture content, such as music (K-pop), dramas (K-drama) and films. The Korea Foundation's global *Hallyu* data from 2015 revealed an estimated 35.59 million fans around the world, a 63% increase from 2014's estimated 21.82 million fans.[3] South Korea's government has funded numerous *Hallyu* entities to promote Korean culture, food, traditional arts, and so forth, overseas (Kang, 2015). The Korea Creative Contents Agency (part of the South Korean Ministry of Culture, Sports and Tourism) estimated that a total revenue of over 60 billion USD for the creative industry sector alone in 2017 (Korea Trade Investment Promotion Agency, 2018). According to the Korea Creative Contents Agency report (2018), cultural content products refer to 'tangible or intangible goods and services, or any combination thereof, which create added economic value based on a system of cultural elements with artistic, entertainment and recreational value, as well as originality and popularity' (p. 5).

The Korean music industry was recognised by *Time* magazine as 'South Korea's Greatest Export'.[4] For instance, Psy's 'Gangnam Style' music video was the first YouTube video to reach 1 billion views, achieving widespread coverage in mainstream media; as of November 2017, the video passed 3 billion views. To put this into perspective, Big Bang – the top act in its niche – took home 44 million USD in pre-tax earnings over the past year, easily more than the 33.5 million USD collected by today's highest-paid American all-male arena pop group, Maroon 5, according to *Forbes*.[5] BTS, an all-boy band, achieved their third no. 1 hit on the Billboard album chart in less than a year (between 2018 and 2019) – The Beatles are the only other band that has been able to get three top sellers in such quick succession.[6] Following the success of BTS in the United States, Blackpink, an all-girl band, saw one of its songs become the most-viewed music video premiere of all time on YouTube with 56.7 million views in its first 24 hours, according to *Variety*.[7] In terms of partnerships, South Korea's top entertainment agency, SM Entertainment, said that it had formed a strategic partnership with Alibaba Group Holding. The Chinese e-commerce giant will buy around 30 USD million worth of SM Entertainment's newly issued shares, gaining a 4% stake in the Kosdaq-listed company.[8]

A strong business ecosystem is formed by an array of stakeholders that all have different interests and roles in making K-pop a success. They include artists, entertainment companies, the media (newspapers, magazines), fans, digital platforms, event managers and firms. Some of the stakeholders have been present in the ecosystem for a long time, while others joined more recently (ie, digital platforms) as the platform incorporated the use of IT to become a business ecosystem. The entertainment industry is generally considered to be self-organising. However, there are a number of focal firms (ie, the content producers) that dictate the rules that govern value creation and sharing.

Table 2 describes the main stakeholder roles present in the K-pop business ecosystem. Content creators are responsible for the creation of the substantive content of work (Greenberg, 2003). The growth of content creation practices in the digital age is viewed as both a political and a cultural emancipatory force and, more pessimistically, as exploitation for the benefit of others (Brake, 2014) (so much so that content creators are sometimes kept away from end-customers by network operators through 'walled garden' portals (Peppard & Rylander, 2006). In many instances, there are calls for systems and methods that allow content creators to protect and profit from the materials they provide and organise (Hong, Tam, & Kim, 2006; Picard, 2000), especially when the new content creators are non-professional users who create and share user-generated content (UGC) in small communities for a limited audience (Obrist, Geerts, Brandtzæg, & Tscheligi, 2008). Moreover, only well-known content creators are often able to secure a content-related agreement (eg, a distribution agreement or licence) with a content provider (eg, a distributor, aggregator, publisher) within the context of the entertainment industry (Brandstetter & Spears, 2011). In this study, we will examine how these stakeholders in the ecosystem develop into boundary spanning actors through the digital transformation of the industry.

3.1.1 | Background of stakeholders in the K-pop ecosystem

In this section, we provide examples of stakeholders highlighted in Table 2 that form the basis of our investigation and their roles in the ecosystem. First, SM Entertainment is one of the three largest entertainment companies (SM, YG, JYP) and content creators in South Korea. SM Entertainment focuses on the production of movies and TV drama content as well as artist development. Its subsidiary, SM C&C, houses and manages actors and idols in their productions. Fans attribute the cause of why much of K-pop music sounds the same to the trend of smaller entertainment companies following in the footsteps of big production houses like SM in an attempt to emulate their success and without experimenting with new genres of music. LOEN Entertainment is another content producer that started as a content broadcasting platform business (Melon) but recently shifted more towards content licensing and artist management lines. Their main competitiveness comes from sourcing from good content and working mainly with content distributors.

KBS is a public broadcaster, and its key service is to provide creative content to the public of Korea. KBS has IT departments dedicated to digital services and digital planning that devise strategies for its long-term development on digital platforms. Kakao is another renowned creative content provider in Korea. Kakao's services, including direct messaging (KakaoTalk), marketplace (KakaoPage) and vehicle hire (KakaoTaxi), are utilised by a significant portion of the Korean public. It launched with its main messenger application, KakaoTalk, in March 2010, hitting 10 million

TABLE 2 Description of the main stakeholder roles in the K-pop business ecosystem

Role	Description	Stakeholder examples	References
Content creator	Responsible for the creation of the intellectual content of a work. Create and add value to content.	SM Entertainment, LOEN Entertainment	Greenberg (2003))
Content provider	Positioned as an intermediary to distribute content in the value chain.	Korean broadcasting service (KBS), KakaoTalk, line messaging service	Farhoomand and Lovelock (2001) in Swatman, Krueger, and Van Der Beek (2006)
Content user	Individuals or groups that consume, interact with, control, create and distribute media content.	Fans of BTS boy band, fans of Seventeen boy band	Daugherty, Eastin, and Bright (2008)

users in its first year.[9] Kakao's IT department comprises engineers with extensive experience in Internet businesses and online gaming. The focus of the company on creative content development and K-pop artist development coincided with the acquisition of Daum web portal and the Loen-Melon merger.[10]

3.2 | Data collection

Data collection began in December 2015 and lasted through July 2018. The purpose of the data collection was to learn from multiple stakeholders in the Korean entertainment industry how a dynamic business ecosystem is developed and organised through IT, such that it promises its consumers a new, seamless way to experience content, innovation and Korean culture. We conducted 30 interviews from which we generated over 50 hours of recordings. The selected informants represented all the major stakeholders of the K-pop ecosystem. Table 3 lists the interviewees and their positions in their organisations, the industry and stakeholders role performed by interviewees and their organisations, the location and language of interviews and an example of the interview topics discussed.

In line with our literature review, we narrowed our focus of inquiry to three pertinent areas during data collection: (a) the evolution of K-pop and Hallyu and the role of digital technologies, (b) business ecosystems and their BP and (c) the development and business value achieved by the focal firms (and/or platforms) and their constituents.

The informants included predominantly senior and middle management at entertainment companies, artist management companies, social media platforms, creative content companies, traditional broadcasting companies and their subsidiaries in Korea. We chose these informants deliberately to leverage the depth of knowledge, experience and leadership (especially in championing IT use) that managers often possess (Bassellier et al., 2003). Informants also included artist managers, event managers, media and fans of K-pop across Asia. We conducted face-to-face interviews at various sites, including offices and concerts in China, Hong Kong, South Korea, Singapore and Australia. Thus, we were able to effectively capture participants' interpretations, illuminate important factors in-depth and follow up with questions for clarification (Oppenheim, 1992; Walsham, 1995). Secondary data sources, including newspaper articles, books, videos and information from corporate websites, were used to supplement our analysis and enhance our understanding of the data collected during the interviews. We also triangulated the data using documents and archival records accessible via online public domains (Neuman, 2010).

3.3 | Data analysis

We analysed the data as they were collected to take full advantage of the flexibility of the case research approach (Eisenhardt, 1989). The voluminous amount of qualitative data was first condensed into a manageable form using narrative and visual mapping strategies (Langley, 1999). The narrative strategy entailed the construction of a 'story' that represented our account of what happened. The visual mapping strategy, on the other hand, involved creating chronological event timelines and documenting our emergent theoretical ideas in a series of diagrammatic sketches (eg, see Figures 1, 2 and 3 in our Findings section). After the narrative and visual maps were constructed, they were verified with the relevant informants. This was to ensure the validity of both our interpretation of the informants' accounts and the theory that we were developing (Pan & Tan, 2011).

Following the development of the narrative and the visual maps, we organised and coded the data collected using the grounded theory techniques of open, axial and selective coding (Strauss & Corbin, 1998). More specifically, open coding was used to assign conceptual labels to data (eg, excerpts from interviews) to create a pool of first-order categories, and axial coding was used to relate the first-order categories to a number of second-order themes (Gioia, Corley, & Hamilton, 2013). The initial set of second-order themes (ie, our theoretical lens; see Pan and Tan (2011)) consisted of the elements of BP (including the role of BO and boundary spanning actors) identified earlier in our

TABLE 3 List of interviewees and topics discussed (December 2015–July 2018)

No.	Interviewee and position	Organisation, industry (stakeholder role)	Location and language of interview	Interview topics
1	Department head	KBS, analogue TV (CP)	South Korea, Korean	Strategies to address hyper-competition in K-pop; strategic investment and relationships among cultural content producers, providers and users; relationship with customers; development of platforms; cross-industry and transnational music and entertainment partnerships; role of IT vendor, etc.
2	CEO	Loen Entertainment (melon, Starship, A-cube, King Kong), music (CC)	South Korea, Korean	
3	Vice CEO	Loen Entertainment (melon, Starship, A-cube, King Kong), music (CC)	South Korea, Korean	
4	CFO	Loen Entertainment, King Kong, music (CC)	South Korea, Korean	
5	Department head	SM Entertainment, music	South Korea, Korean	
6	CEO and president	AKA TV (K-pop contents production), digital TV (CC)	South Korea, Korean	
7	CEO	JoongAng Ilbo digital platform, media (CP)	South Korea, English	
8	Vice president communication group/ PR and marketing	Naver, Inc. Corp., internet content services (CP)	South Korea, English	
9	Deputy general manager/ PR	Naver, Inc. Corp., internet content services (CP)	South Korea, English	
10	Former department head	CJ E&M, Samsung electronics, TV/movies (CP)	South Korea, English	
11	Vice CEO	King Kong Entertainment, artists management (CC)	South Korea, Korean	
12	Content biz team director	Kakao Corp., internet services (CP)	South Korea, English	
13	Director	Champion entertainment Australia, events management (CC)	Australia, English	Local and overseas music and entertainment partnerships; role of IT vendor; role of IT in fan engagement; local events coordination; role of IT in promotion of *Hallyu* and K-pop, etc.
14	Communications/media manager	Champion entertainment Australia, events management (CC)	Australia, English	
15	Fan 1 – Seventeen concert	Seventeen concert in Australia, fans (CU)	Australia, English	
16	Fan 2 – Seventeen concert	Seventeen concert in Australia, fans (CU)	Australia, English	
17	Content consultant – general entertainment	Turner international Asia Pacific, Ltd., events management (CC)	Australia, English	
18	Editor-in-chief	Helloasia.com.au, media (CU)	Australia, English	
19	Entertainment correspondent	Korean Foundation for International Culture Exchange, media (CP)	Australia, English	

(Continues)

TABLE 3 (Continued)

No.	Interviewee and position	Organisation, industry (stakeholder role)	Location and language of interview	Interview topics
20	Managing director	Culture content industry agency Jeollabukdo, governmental agency (CP)	South Korea, Korean	Role of governmental agency in promoting Korean content industry; cultural agency and industry working relationships; role of technology in delivering cultural content, etc.
21	President	Culture content industry agency Jeollabukdo, governmental agency (CP)	South Korea, Korean	Role of 3-D printing technology in cultural content delivery; plant-based products production, etc.
22	Manager	Culture content industry agency Jeollabukdo, governmental agency (CP)	South Korea, Korean	
23	Manager	3-D printing, governmental agency (CP)	South Korea, Korean	
24	CEO	(Twintech) virtual simulation and production, technology (CP)	South Korea, Korean	Cultural content production equipment, mock instruments and cockpit, etc; role of 3-D games and virtual reality games; experience in cultural content delivery, etc.
25	Executive director	Gaming business (Moazio), gaming (CC)	South Korea, English	
26	Public relations manager	Investor, Lccmall.com, venture capitalist (CP)	China, English	Investment and relationships between cultural content producers and platform, etc.
27	Executive director	Investor, Lccmall.com, venture capitalist (CP)	China, English	
28	Manager	Music and entertainment partnerships twitter Korea, internet services (CP)	South Korea, English	Music and entertainment partnerships, role of IT vendor, etc.
29	Entrepreneur and director – two interviews	Music, games and video, entertainment (CP)	Singapore and China, English	Cross-industry and transnational music and entertainment partnerships; role of IT vendor, etc.
30	Director	Hong Kong Cyberport, internet services (CP)	Hong Kong, English	IT support for start-ups and communities; music, games and video entertainment; creative digital clusters

literature review. The second-order themes were then abstracted into three aggregate dimensions that emerged based on the evolutionary trajectory of the K-pop ecosystem across three distinct periods (ie, production, platformisation, democratisation; to be elaborated in Section 4). If we were confronted with data that challenged or did not easily fit into our existing themes or dimensions, then we modified our coding scheme accordingly (ie, we added a new, or modified/deleted an existing, second-order theme or aggregate dimension; see Strauss and Corbin (1998)), and coding was restarted. By 'recursively iterating between (and thus constantly comparing) theory and data' (Eisenhardt & Graebner, 2007) in this way, our theory was inductively derived and gradually shaped. This process continued until the state of theoretical saturation was reached (Glaser & Strauss, 1967), which refers to the state in which the inductively derived model can comprehensively account for the case data and 'incremental learning is minimal because the researchers are observing phenomena seen before' (Eisenhardt, 1989). The coding and organisation of data is illustrated in Tables 4 and 5 in Section 4.

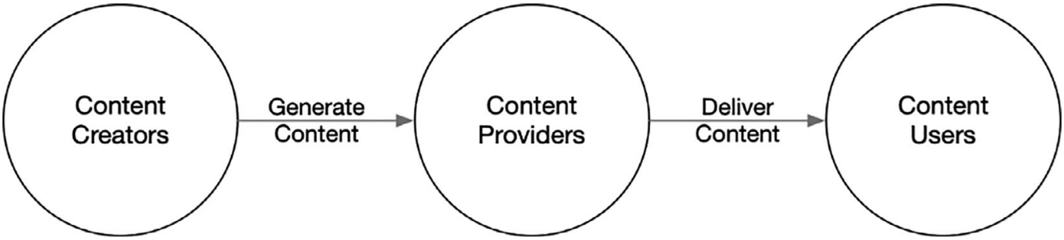


FIGURE 1 Linear value chain in content delivery

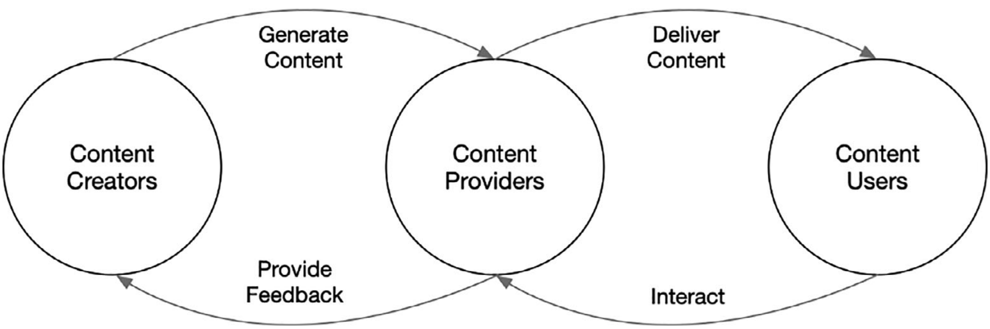
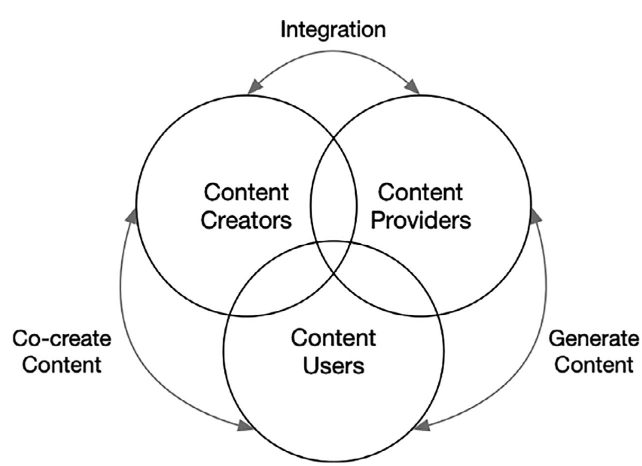


FIGURE 2 Inception of digital platforms for content delivery and value

FIGURE 3 Democratising content delivery and value network



4 | CASE FINDINGS

4.1 | Linear production and delivery of content and value (1990–2000)

From the mid-1990s, consumers across East Asia were confronted with issues of accessibility to new digital pop content and related content services (Dator & Seo, 2004). It is noteworthy that during this time, much of East Asia was

TABLE 4 Key boundary practices and the role of IT in platformisation

Data to first-order concepts	Second-order themes	Aggregate dimensions
Technology platforms fractionated benefits of content creators '*Whilst monetisation benefited the distribution platforms, it did not achieve the same results for the content production companies themselves'* – CFO, LOEN Entertainment (CC) **Technology platforms provided unregulated access to content** *'The physical albums market reduced … also largely due to piracy on XXX (removed for privacy purposes)'* – Vice-CEO, LOEN Entertainment	Platforms relinquish control of content creators on content production and distribution	Platformisation of the music and entertainment industry includes the development of an identity, reputation and business models
Technology enables shared rights '*Our good collaborative relationships allow low prices to be paid to the licenser to ensure some revenue, and working together to protect the copyrights'* – Vice-CEO, LEON Entertainment **Technologies as change agents** *'we have successfully transformed the old analogue music provision to a modernised platform-based service (BO), unlike Japan*' – Department head, KBS TV	On-demand platform-based services embody new pricing and incentive models to protect content providers and consumers	
Technology to increase reach *'If we did not have these channels, we could not reach other countries, in the traditional media market*' – Head of public relations, SM Entertainment '*Webpages (BO) were created by fans (BS) to fill in the gap as their first stop, before even looking up official sources such as videos and notices'* – Department head, KBS TV **Technology for market estimation** *'IT allowed for a much easier approximation of local market interest'* – co-CEO, King Kong Entertainment	Platforms generate new markets, connections, collective resources and value pooling with content consumers	

Note: BP – 'Boundary practices' are a set of activities in which IT is incorporated so that a variety of firms may execute so as to either reinforce or help bridge existing boundaries within and across organisations; BO – 'Boundary objects' refer to artefacts used by actors to acquire both local usefulness (which actors jointly recognise and value) and a common identity in practice; BS – 'Boundary spanners' refer to organisational actors who establish the local usefulness of BO-in-use and establish their common identity (CC, CP and CU).

facing a period of financial crisis (Kim & Vasileva, 2017). Many traditional companies were unable to make technological investments. Music was analogue, distribution occurred via physical channels and production was based on large scales and budgets. In the Korean entertainment industry, official sources and broadcasters could not provide comprehensive directories to profiles, photos, videos and extra behind-the-scenes footage. It was such an issue that, as the head of the analogue department at KBS explained, demand grew for '*web pages [that] were created by fans to fill in the gap as their first stop, before even looking up official sources such as videos and notices*'.

TABLE 5 Key boundary practices and the role of IT in democratisation phase

Data to first-order concepts	Second-order themes	Aggregate dimensions
Technology was complementary to the linear, controlled distribution of content *Content providers select the content they want and believe is most suitable to deliver to content users* (CP). **Consistent content production** Large entertainment companies like SM Entertainment have a reputation of 'factory packaging' their artists (BO), often restricting their artistic freedom and limiting the inclusion of their self-composed songs in albums.	Content creators and providers sponsor infrastructure and exercise rules through platforms in order to stylise content	Democratisation of music and entertainment industry includes the development of co-creation, new interdependencies and mass innovation
Technology determines conversion *'They can successfully persuade stakeholders (BS) with intellectual rights (BO) to convert over to a packaged pricing marketing strategy, the first in the world … Overall, Korea has been quite innovative in terms of policy-making and technology (BO) since the early 2000s'* – CEO LOEN Entertainment	Content creators and providers refine policies on platform engagement and use	
Technology allows consumer participation in content production *'Recently, Naver has released the V LIVE App, and now when they have a V LIVE broadcast, after a few minutes or hours, the video gets re-uploaded with subtitles. Recently, they [agencies] have been using the fans' subtitles (BO), so they are actually reaching out to fans from different countries who can translate Korean to their language, and that's on the official App. (BO)'* – Administrator, BTS Army Fanclub **Non-economic exchanges** *'To them [artist agencies] KakaoTalk is just one additional channel … I do not think we had any economic exchanges; we did not pay money to them and they did not pay money to us. It was a benefit to them and a benefit for us'* – Former executive, Kakao	Non-contractual value co-creation between content providers and content consumers grow	
New technologies *'We are preparing for many technology-related initiatives. We have a complex cultural space called Artium, where we produce hologram concerts. So even if the artists are not present, we can hold concerts using holograms. For VR, we are shooting and showing EXO's [boyband] concerts from 360°. Next, we have things related to IoT'* – New media executive, SM Entertainment *'When we launched a feature within KakaoTalk called Plus Friends, we put in a lot of K-Pop material … For Southeast Asia, we put in SM*	Continuous technological innovations for co-creation	

(Continues)

TABLE 5 (Continued)

Data to first-order concepts	Second-order themes	Aggregate dimensions
Town and SNSD [girl group]' – content director, SM Entertainment		

Note: BP – 'Boundary practices' are a set of activities in which IT is incorporated so that a variety of firms may execute so as to either reinforce or help bridge existing boundaries within and across organisations; BO – 'Boundary objects' refer to artefacts used by actors to acquire both local usefulness (which actors jointly recognise and value) and a common identity in practice; BS – 'Boundary spanners' refer to organisational actors who establish the local usefulness of BO-in-use and establish their common identity (CC, CP and CU).

As such, the creation and distribution model was mostly linear. Content creators produced the content. Meanwhile, distribution was the business of providers such as traditional media channels (ie, TV, radio) and offline stores for physical media (eg, tapes, CDs, DVDs). KBS is an example of a public TV station, and its key service is to broadcast content. It consists of teams that edit short clips out of full episodes of shows and IT departments dedicated to digital services. Furthermore, KBS regularly looks over the trends in the market and trial content if it has been proven to be marketable. In other words, the onus for innovation was exclusively on the content creators. Due to the intermediation of content distribution and lack of direct access to content users, content creators innovate here based on what they think consumers want. The top-down innovation process is unidirectional in this phase.

Content providers would select the content they want to be delivered to the content users. Therefore, the content creators' mission was to determine what content was most suitable for broadcast. Monetisation opportunities within the entertainment ecosystem were limited. The distribution of K-pop content primarily came in two ways: through the music and the artist. While the former dealt with the more direct and traditional forms of distribution through CDs or cassette tapes, the latter depended on live performances and celebrity power to build their brand. As the CFO of LOEN Entertainment (a leading record company in South Korea, now called Kakao M) explained:

> 'Whilst monetisation benefited the distribution platforms, it did not achieve the same results for the content production companies themselves … Entertainment companies had to turn to concerts and exhibition-type performances to gain revenue'.

4.2 | Platformisation as boundary practice (late 2000–2010)

K-pop started to receive global exposure in the early 2000s. One strategy of content creators was to tailor the content to foreign markets **[Boundary Practice]** to generate more revenue. The manager of music and entertainment partnerships at Twitter Korea explained that '*when they targeted the Japanese market, BoA (a Korean musician) was specifically trained for the Japan market, learning Japanese and signing with a Japan-based agency. The type of music she sang resonated more with the Japanese at that time*'. There was also a brief period during which major entertainment companies attempted to penetrate the US market with different groups or single singers: JYP with Wonder Girls, SM with BoA and YG with one of 2NE1's singers, namely CL. At the same time, the companies were aware that this situation was not viable in the long run because their investments were unable to generate significant returns.

Exacerbating the situation was the fact that the emergence of broadband and digital distribution platforms in the early 2000s caused a steep decline in the traditional media market. Changing consumer preferences required entertainment companies to improve their interface with their customers. The CEO at LOEN commented that '*the physical albums market reduced to 10% of its original size within five years*'. The emergence of broadband also meant that entertainment companies and broadcasters (both content creators and providers) had to deal with piracy. As the Vice-CEO of LOEN and Melon entertainment explained:

> 'The physical albums market reduced … largely due to piracy on XXX (name removed for anonymity). In 2007, the court ruled that piracy is illegal and officially declared XXX as such. The site transformed into a legal, paying site in 2008 but still faces high market competition to this day'.

Content creators were facing stiff challenges interfacing with their users with legitimate content, due, in part, to a lack of formal agreements with broadcasters concerning the provision of licensed content to consumers. This meant that lower prices made licensers more profitable. However, as K-pop grew, the concentration and disproportion of the distribution of returns appeared, and the industry had to change its model to adapt to these changes. Large entertainment companies such as SM Entertainment had a reputation of 'factory packaging' its artists, often restricting their artistic freedom and limiting the inclusion of their self-composed songs in albums, preferring instead to go with established, famous producers. In other words, content creators had to partner more closely with content providers to innovate. Once platforms and communities were established, competition emerged around the production of content by creators. As the CEO of LOEN detailed:

> 'It was more for survival that we built K-pop. We had to produce high quality of video in a short time, while CDs (Compact Discs) became collector items. The requirements to succeed became different…'.

The new practices by digital platforms shaped the emergence of new knowledge, artefacts and communities of practice (Venters & Wood, 2007) within the growing business ecosystem, challenging the incumbent technologies and associated socio-technical norms. Traditional broadcasting companies such as KBS sought opportunities to diversify to develop business ecosystems, and their managers explored new digital infrastructures to reinforce established networks by partnering **[Boundary Practice]** with platform businesses to explore and enact emergent complementary functions (Pierce, 2009).

For example, KBS World, consisting of a paid channel and website respectively introduced in 2003 and 2008, was launched to deliver its entertainment content to an international audience. Some of KBS' broadcasted content also inspired the independent and creative market to come up with new, informal content without copyright problems. It encouraged independent content producers to experiment with revolutionary content using different platforms. Even creators and directors considered other digital platforms in the planning and production stages to explore their possibilities.

Here, it is important to establish that Twitter and Facebook **[Boundary Objects]** became the main platforms for fans to share information and news about artists, including behind-the-scenes footage for weekly music shows, fansubbed material and images taken by fans. Notably, Korean became the seventh language that Twitter supported from 2010, when the social media platform partnered with Korean web portal Daum to display Korean tweets. According to the manager of music and partnerships at Twitter Korea, '*Twitter has helped many big bands including EXO and Big Bang to connect with their fans on upcoming album launches, generating interest and tweets*'.

Figure 2 illustrates how content creators and content providers **[Boundary Spanners]** within the business ecosystem drew closer to one another in this new phase as the locus of innovation expanded to include both entities.

The department head of KBS TV commented that 'we *have successfully transformed the old analogue music provision to a modernised platform-based service, unlike Japan*'. In addition to more K-pop music content, content providers also shaped content to reflect changes in the delivery of music content in the increasingly digital and urbanized landscape of South Korea. New digital platforms were created by a collaboration between big free to air broadcasters. For example, POOQ, a subscription based on-demand service that allows viewers to watch TV programmes on their mobile devices, is a joint venture of three of Korea's free to air broadcasters: KBS, MBC (Munhwa Broadcasting Corporation) and SBS (Seoul Broadcasting System). Furthermore, newer K-pop artist groups tended to be more technology savvy and connected, while older groups tended not to have dedicated applications for their fans.

4.2.1 | Overcrowding platforms, pricing issues, monetisation

The excessive costs paid by users for content received through telecommunications (eg, through text messaging) also provided a business opportunity for new, available-for-free messaging platforms. One of the key platforms was Kakao, under which the application KakaoTalk was launched in 2010. KakaoTalk rapidly grew to become the de facto messaging application in Korea. KakaoTalk recognised that iOS and Android devices would change the content business landscape. Initially, these devices did not offer voice and video communication options, as telecommunication companies already offered them. Kakao's starting point of adding value through free text messaging led to an expansion in the areas of games, products and brands. Here, it is helpful to note that Kakao's former CEO commented that '*the driving force of Kakao is K-pop*'. The application's integration of K-pop content was a unique strategy that utilised the popularity of K-pop to develop its brand. For example, Kakao had a 'Plus Friends' feature that allowed users to befriend brands, including K-pop artists and entertainment companies, to receive content directly from those entities (see Figure 4). While competing messenger services, including LINE and WeChat, were operating in the market at the same time, KakaoTalk stood out as the application that heavily marketed K-pop artists.

Consequently, K-pop became a way to monetise the platform. The director of the content business team at Kakao explained that '*commercial contracts were made for Plus Friends partners. The messages sent were considered to be a marketing tool for the partner. Hence, Kakao found a way to profit from them*'. Another way to monetise activities on KakaoTalk was to sell sticker and emoticon packs to enrich instant messaging. In particular, Kakao's knowledge and experience with the Korean market allowed it to introduce relevant K-pop emojis and stickers. Without these new means of engaging with the market, Kakao's content business director commented, Kakao's '*US-based service would have not been able to gain as much traction*'.

Since 2007, another important change has happened: major shifts have occurred in the contract terms between licensers and content providers in Korea. As LOEN's CEO explained:

> 'Overall, Korea has been quite innovative in terms of policy-making and technology since the early 2000s. They are able to successfully persuade stakeholders with intellectual rights to convert over to a packaged pricing marketing strategy, the first in the world. The price allows the user to download a specific number of songs over a period'.

The pricing decisions made as part of the scheme, in particular, have brought about further changes to the digital content market. Due to piracy in the music market, unless the licenser and platform providers are able to collaborate, the licenser may lose out as they are affected by illegal downloads. The licensers and platform providers have therefore in many cases been vertically integrated. LOEN now had to operate more as a licenser than a platform provider; this transformation was brought about as the market shifted into the digital age. As LOEN's Vice-CEO further explained:

> 'Our good collaborative relationships allow low fees to be paid to the licenser to ensure some revenue, then working together to protect their copyrights. Consumer perception changed initially when they had to start paying for music, and gradually the price was increased to allow the licenser to gain profitability. When collaborations first began, the prices were at their lowest in the market. Now, it has one of the highest competitive price levels in the world'.

Traditionally, distributors took a large cut of the revenue generated. As a result, many entertainment companies were not profitable. Looking for alternate sources of revenue through globalisation was equally difficult because they were unable to estimate the market size and potential. The developments of digital infrastructure not only created new interdependencies (Caglio & Ditillo, 2012; Gerdin, 2005; Macintosh & Daft, 1987) between content providers and content creators but also presented a means to support product and service diversification. While initially this was a survival tactic for the Korean market, the high quality of the artists also caught the attention of other countries, triggering the globalisation of K-pop content. Hence, IT was seen as one of the key enablers for the globalisation of Korean entertainment products. Companies just had to do what they have always been doing, but they would now be digitally connected to online platforms, accessing international audiences and enabling them to better achieve their market potential. As the co-CEO of King Kong Entertainment, a talent agency, explained:

> 'Before, it was hard for companies to explore the viability of holding events overseas. IT allowed for a much easier approximation of local market interest. The data provided through social media can be so detailed they are able to make precise decisions to start new projects. It is no longer the case that they have to personally conduct studies in foreign markets to create specifically catered content; what they are doing locally, if done well, can appeal to overseas markets the same way it does locally'.

Overall, the role of IT was critical to the global spread of K-pop groups. High-quality content (eg, music videos) was typically first released through YouTube, then spread and shared through various social media platforms. Because of content sharing among international communities of users and fans of the various groups (generating network effects; see Bharadwaj et al., 2013), certain artists performed especially well in various markets. The co-CEO of King Kong Entertainment added:

> 'Some (artists) are popular in Korea, while others are popular outside of Korea … Certain markets are more systematic, some are more chaotic, but writers and producers have the biggest potential to […] draw powerful distributors in their local market'.

Digital platforms went further and became pro-active in creating exclusive and customised content for different countries. For example, Kakao's marketing strategy in South-east Asian countries was to couple a local artist (such as Sherina Munaf in Indonesia) with a Korean artist (such as Big Bang). In other words, content producers were trying to foster customer interactions across the globe. The head of the Artist Public Relations Team at SM Entertainment commented on the significance of digital channels for globalisation:

> 'We have 45 million subscribers on Facebook, 8 million on YouTube, KakaoTalk – 9.2 million, EXO-L – 3.2 million, and LINE – 43.3 million … We use YouTube **[Boundary Object]** for music videos but on other channels, we upload photos and news. In the case of EXO-L, we made an app dedicated to the fan club of EXO [idol group] … If we didn't have these channels, we could not reach other countries, in the traditional media market we couldn't … Right now, we only put them [content] on the Internet, then we can promote them in Taiwan and Japan simultaneously. But before it was very difficult so we needed promoters in other countries'.

Table 4 summarises how we organised and coded the data collected to form emergent BP, BO, the role of IT and boundary spanning actors in the platformisation phase.

4.3 | Democratisation as boundary practice (early 2010 onwards)

In this third phase, the priority of focal entertainment companies shifted to fostering symbiotic relationships with other members of the K-pop ecosystem. The two previous stages focused on attracting potential members to join the ecosystem and strengthening their relationships. This phase of ecosystem development sought to further expand and strengthen global partnerships and to encourage ecosystem members to collaborate as a single community (Madlberger & Roztocki, 2009).

This phase was essential in extending benefits to the ecosystem's peripheral members (Linden, Kraemer, & Dedrick, 2009) and created a synergistic impact on the ecosystem's overall performance (Li, 2009; Tan et al., 2015). This shift to co-creation recognises the autonomous contributions of both producers and consumers towards innovation efforts and redefines the division of labour between producers and consumers through their activities (Zwass, 2010). Because even consumers become co-creators of value and content, we consider this transformation the democratisation of innovation (Eaton, Elaluf-Calderwood, Sorensen, & Yoo, 2015; von Hippel, 2005). Here, we use 'democratization' (per von Hippel (2005)) to characterise the changes in traditional product innovation and consumer participation. Technology underscores radical and rapid increases in consumers' abilities to innovate, and businesses tap the creativity of consumers for product innovation (von Hippel, 2005).

As co-creation takes place, the different actors form a much more integrated and interactive ecosystem (see Figure 3), and their roles, in particular, tend to converge. Content creators integrate with content providers to distribute content more efficiently. Meanwhile, content users become more involved in content creation. As a result, interdependencies increase and the locus of innovation is extended to include all actors.

In the past, content users (ie, fans) only consumed content. With the rise of digital platforms, entertainment companies provided incentives for fans to become more involved within the ecosystem. For example, there is a growing trend for live survival-style audition programmes (Oh & Lee, 2014). JYP's *'Sixteen'* and CJ E&M's *'Produce 101'* are examples of this trend. In these programmes, trainees compete with one another for a spot in an upcoming idol group. The fans play a key role because they are responsible for selecting the members for prospective idol groups through an online voting system on the companies' platforms. The fan groups also participate in content creation. Accordingly, while broadcasting companies used to provide digital content with subtitles, they have now pushed this task to the fans. Entertainment agencies are increasingly encouraging and becoming reliant on content creation and circulation via social media driven by the fans. This is explained by an administrator of AusArmy Project, an Australian fanbase for the boyband BTS:

> 'With the creation of viral videos **[Boundary Objects]**, the fans show their love for the idols they adore. So if they find a two second bit of a video that they find so funny or relatable, they would share that via some sort of means, with GIFs or photos with captions. There are also a lot of fans who take original content and put subtitles, and that allows a bigger spread … Recently, Naver has released the V LIVE App, and now when they have a V LIVE broadcast, after a few minutes or hours, the video gets re-uploaded with subtitles. Recently, they [agencies] have been using the fans' subtitles, so they are actually reaching out to fans from different countries who can translate Korean to their language, and that's on the official App'.

The contributions of fans have become so significant that entertainment companies are creating new business models specifically designed for the active involvement of content users **[Boundary Spanners]**. On top of the online channels used for the distribution of their K-pop artists' content, entertainment companies are also establishing new external platforms (Gawer & Cusumano, 2014) designed for UGC. Consumers and artists can create and host their own shows there, instead of the conventional engineered content constructed and owned by professional content producers (Kim, Na, & Ryu, 2007). A former Content Licensing Coordinator at CJ E&M described their own UGC platform:

> 'CJ has their brand called DIA TV … DIA TV is a digital platform for individual players. They call them [users] the content creators in recent years. They could be ordinary people or actors who do their own show. I believe it's going to be a very interesting market … YouTube is a platform, but MCM (multi-channel network) I would say is sort of management'.

A New Digital Business Ecosystem: A New Role for IT.

During a keynote in 2016, SM Entertainment announced the launch of several digital platforms to increase interactions within the K-pop ecosystem. Soo Man Lee, CEO of SM Entertainment, presented an update on a new strategy that he termed ‘New Cultural Technology’. The first project launched was the SM STATION, which digitally released one music video and its associated audio every week. A new label called EDM would act as a platform to create more collaborations with external partners. A department head of SM Entertainment and music explained:

> ‘Though most of the projects involved its in-house artists, there were some collaborations as well as inviting guest producers who normally would not participate in SM productions’.

In addition, three other platforms were launched to create a 24/7 digital playground for K-pop fans. For example, everysing is a platform that enables anyone to sing along to a high-quality live accompaniment and record the song for free. It is even possible to sing a duet with a celebrity of the consumer's choice (see Figure 5).

Another platform, everyshot, was launched to let fans create their own music videos. Meanwhile, Vyrl is a photo SNS based on a matter of interest. This app lets fans access a large database of celebrity photos, get news and gossip about their favourite idols, participate in global chats and meetings with stars and access live concerts through video streaming service. Rookies Entertainment (Figure 6), on the other hand, is an innovative platform in the K-pop ecosystem. Fans can become producers using a mobile application. They can experience the whole production process, receive feedback on their management skills and get credited on the debut album of their group. If a fan is doing the job particularly well, an award and a potential internship can be offered. SM Entertainment also invested in multichannel network. The idea was to create visual radio broadcasts, web dramas and other forms of web entertainment (eg, sports lessons, weight training) with SM celebrities. The content will be offered globally without border restrictions.

Entertainment companies are also innovating with new media content that is developed through collaboration with external technology companies. The director of New Media Division at SM Entertainment elaborated:

> ‘We are preparing for many technology-related initiatives. We have a complex cultural space called Artium, where we produce hologram concerts. So even if the artists are not present, we can hold concerts using holograms. For VR, we are shooting and showing EXO's [boyband] concerts from 360 degrees. Next, we have things related to IoT. So we have many partners such as Samsung and Google. We have a department that plans for VR [content] but we work with technology companies who can produce them. Technology companies need our context so we collaborate’.

Technology is also being utilised to a greater extent for the distribution and promotion of K-pop content. At the same time, however, it is difficult to monetise the success of those channels. Again, globalisation seemed to be an attractive way to generate more revenue. However, K-pop faced high barriers in penetrating foreign markets such as North America. Therefore, entertainment companies have established alliances with a number of international partners who are based overseas to further expand their operations globally. While these partnerships were not very lucrative, they did allow K-pop to gain more exposure. As the head of the Artist Public Relations Team at SM Entertainment explained:

> ‘We provide [content] to Apple's iTunes and Google because we have no [global] channel. Because in terms of the global [market], it is important to prioritise popularity first. The revenue from iTunes is pretty good as well, but gaining popularity is more important because there are other things like concerts that we can monetise more than streaming music’.

Interestingly, K-pop also contributes to the global expansion of various Korean social media platforms, such as Naver Corporation's LINE and KakaoTalk. These mobile applications were able to successfully penetrate and attract a large

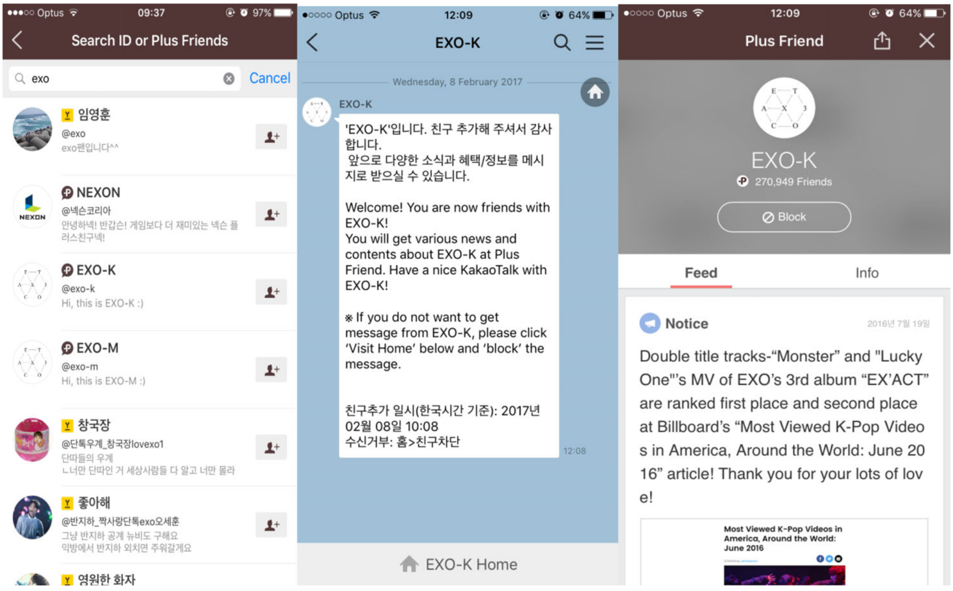


FIGURE 4 Kakao's Plus Friends feature

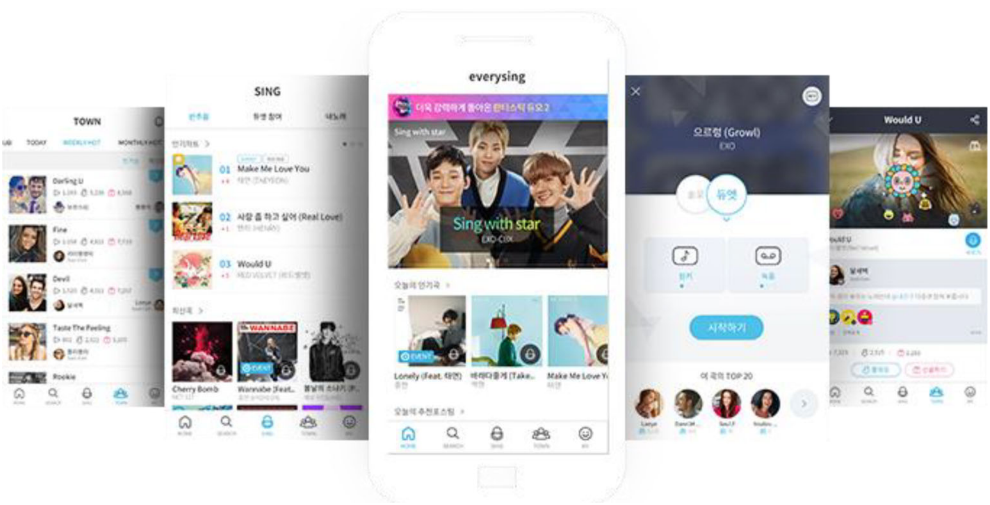


FIGURE 5 everysing, a smart karaoke application by SM Entertainment

number of initial users in foreign markets by collaborating with Korean entertainment companies. For example, the former CEO of Kakao described the workings of their partnerships with entertainment content providers:

> 'When we launched a feature within KakaoTalk called Plus Friends, we put in a lot of K-Pop material … For South-east Asia, we put in SM Town and SNSD [girl group], different K-Pop stars or their management companies, so if you befriend the star or the management company, you get pictures, information about new releases and things of that nature, and that really helped us gain users … To them

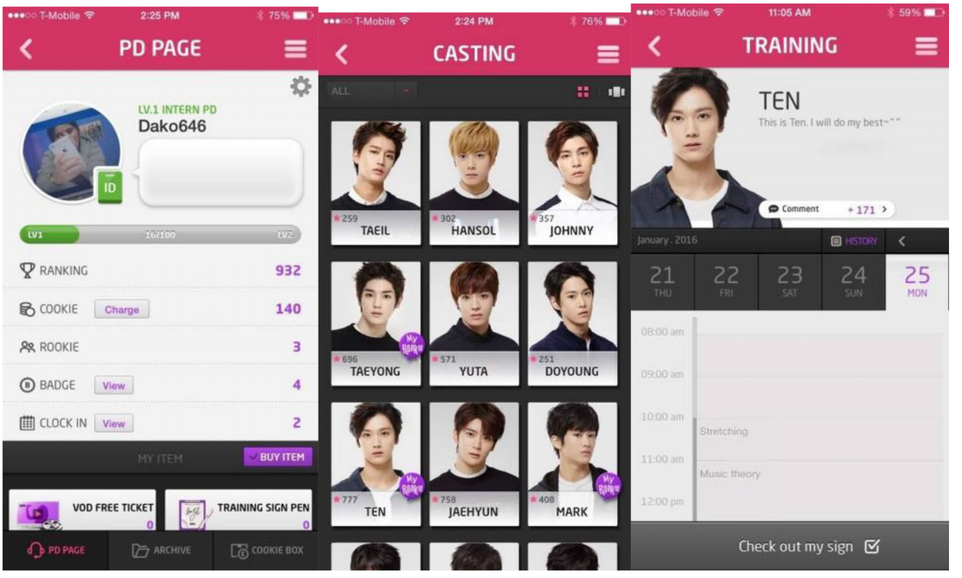


FIGURE 6 Rookies Entertainment mobile application

> [artist agencies] KakaoTalk is just one additional channel … I don't think we had any economic exchanges; we didn't pay money to them and they didn't pay money to us. It was a benefit to them and a benefit for us. We did have a written agreement for the Plus Friends partnership but this was not a commercial deal'.

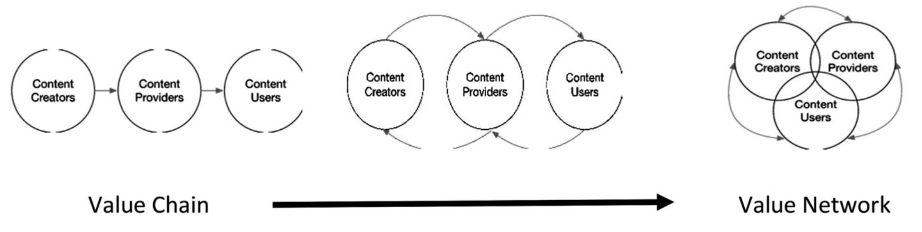
Consequently, all the stakeholders of the K-pop ecosystem developed further interdependent relationships. The partnerships with large technological companies empowered the entertainment agencies to transcend previous organisational boundaries (Cash & Konsynski, 1985) that were once circumscribed within the music industry. Entertainment companies, technology partners and fans evolve simultaneously, facilitating the co-evolution of entities within the entire K-pop ecosystem. As a result, a symbiotic ecosystem formed with different members that closely collaborate as a single entity towards the shared purpose (Brusoni & Prencipe, 2013; Li, 2009; Tan, Pan, Lu, & Huang, 2009) of promoting K-pop.

Table 5 summarises how we organised and coded the data collected to form emergent BP, BO, the role of IT and boundary spanning actors in the democratisation phase.

5 | DISCUSSION

Based on our longitudinal investigation, we derived a framework from our findings that captures not only the digital transformation of the Korean entertainment ecosystem but also the BP, the dominant IT artefacts and their formative roles in developing the business ecosystem. Through this empirical investigation, we addressed our research question and sought to better understand the role of IT in enabling effective BP in the development of a business ecosystem. We also shed light on the role of IT in enabling effective content generation and delivery in the context of the global K-pop entertainment ecosystem. In our framework (see Table 6), platformisation and democratisation capture key BP and salient uses of IT that drive the transformation of content delivery from a value chain to a value network. The role of IT is essential in providing the ability for the different boundary spanning actors to collaborate

TABLE 6 A framework for the development of business ecosystems

Boundary practice and objective	Production Traditional content production and delivery	Platformisation Development of an identity, a reputation and business models	Democratisation Co-creation, new interdependencies and mass innovation
Formative role of dominant IT artefact	IT to alter and stylise content production: emergence of broadband put pressure on content creators to produce digital content.	IT to create and share distribution channels: IT helps to connect the different actors. Strong role of digital platforms to distribute the content. Digital content is adapted to platforms. Platforms help to create suitable content for users.	IT to reshape and foster new capabilities: social media, networks and mobile applications help to connect all the stakeholders. IT enables multi-directional communications among fans, content providers and content producers. IT helps to distribute content globally by offering new channels and translation capabilities.
Impact on business ecosystem	Emergence of content identity: emergence of distinctive K-pop content and style are delivered through large production houses.	Deepening of interdependencies: IT enables market sharing and the development of new content, new markets and new collaborations among creators and providers.	Attainment of self-organisation: reciprocal interdependencies among creators, providers and users create a symbiotic and dynamic self-organising content creation ecosystem with multiple channels for delivering and consuming content.
Illustration of content delivery in ecosystem			
	Value chain		Value network

and converge in their offerings of digital innovations in content delivery. The role of IT is therefore not only supportive but formative, and critical to the development of new practices, new objects and artefacts and new communities. Interactions become richer and more extensive as technology becomes more social and widely available across the K-pop ecosystem.

During the initial stages of the ecosystem, technologies like optical disc drives and broadband internet support upstream content producers and shape the mechanisms of interaction with downstream stakeholders (Rapp, Beitelspacher, Grewal, & Hughes, 2013). In effect, this fosters the delivery of generated content in a linear value chain. BP are primarily enacted by upstream stakeholders such as large production houses that become responsible to produce a distinctive style of content. Downstream stakeholders, such as media platforms and artists, are limited in their activities because their roles are well-defined. IT and practices, therefore, determine the identity of stakeholders and the rules of operating in the industry (Tsatsou, Elaluf-Calderwood, & Liebenau, 2010). Hence, content creators become responsible for the identity of the business ecosystem and determine its creativity, style and power. In this ecosystem form, content distribution among stakeholders is mainly offline, while online activities are minimal and of a linear form; this creates predictable business models.

In this study, we find that the emergence of broadband internet and the growing concerns of content piracy coerces content creators and providers to transform their business models. Hence, more contemporary digital platforms emerge to form new distribution channels. Moreover, the sharing of assets in telecommunications, licensing practices, IT capabilities such as messaging services, the convergence of technology and interests amongst the actors in the ecosystem increases. In our research context, the onset of asset co-specialisation, which refers to when assets and capabilities are shared to generate business value (Santoro & McGill, 2005; Tippins & Sohi, 2003), is evident in the collaborations and innovations between the content creators and providers. IT platforms (such as Kakao Plus Friends) and new practices developed through a collaboration among the upstream entertainment companies and their artists and downstream social media platforms foster the development of new content, interdependencies (Kumar & Van Dissel, 1996) and communities (Lindberg, Berente, Gaskin, & Lyytinen, 2016). Platforms like YouTube also lower language barriers in reaching global markets, enabling K-pop to build communities of content users and fans across Asia, Europe, South America and increasingly North America (Chang & Choi, 2011). Platformisation, which refers to the establishment of new digital platforms, is a key aspect of the business' ecosystem development. Platforms enable all actors to interact with less friction. Content creators can easily reach users by making their content available on a platform and users have centralised access to content. Furthermore, influence and creativity are shared by content creators and providers.

During the democratisation phase, convergence accelerates and the effects of co-specialisation increase. Hence, IT fosters a process of self-organisation (Coleman & Henry, 1999) wherein a pattern and regularity of creative content development and innovation emerges from stakeholder interactions and without the intervention of a leader or core entity within the ecosystem. This democratisation of innovation builds on the well-established partnerships content producers have with their providers and users, as highlighted in the previous phases. Their respective ecosystems, processes and communities tend to overlap and merge to such an extent that capabilities and resources are extensively shared. In our research context, entertainment companies, technology partners and fans appear to have a shared purpose (Brusoni & Prencipe, 2013; Li, 2009) and simultaneously contribute actively to new technologies, communities and practices within the entire K-pop ecosystem. Just as production and platformisation require some orchestration from focal firms, democratisation is self-organising in nature, with the different entities working together in an emergent manner, such that the identity of stakeholders (Tsatsou et al., 2010) and the boundaries within the ecosystem (Wang, 2009) become less clear. The relationships are dyadic or triadic, depending on the content exchanged. A true digital business ecosystem emerges with interdependent actors that complement each other to fulfil a common objective.

Organising distributed creativity in ecosystems establishes an open networked community. There is no permanent need for centralised or distributed control. The vertical integration of content, platforms and applications creates digital infrastructures and standardised processes used by all members of the community. Although IT capabilities may function autonomously, the role of each of the platforms is not just a mere orchestrator. Instead, the platforms create a conducive environment and support a process that deliberately melds one activity with others. When tensions arise between content providers and creators, actors from diverse ecosystems introduce new mechanisms to overcome difficulties in gaining alignment rather than compel actors to demonstrate commitment, giving rise to self-organisation (McKelvey, 2016) and mass innovation (Phelps, 2013; Van Osch & Avital, 2009). The blending of operational principles and their logics becomes a very comprehensive effort, tending to sophisticated demands. Deeper interdependencies drive co-creation and innovation among supply-side businesses and demand-side consumers. As a result, no external or internal leader sets goals or controls the ecosystem, but the events occur spontaneously from interactions in time, tending to self-organisation.

5.1 | Implications for theory and practice

Based on the above discussion of our research findings, we now turn to the implications of our study for established theory and literature on business ecosystems, BP and interdependencies and IT-enabled creative processes.

5.1.1 | Business ecosystems and platforms

Our empirical study contributes directly to the emergent body of knowledge on the development of platform and business ecosystems (Majchrzak, Markus, & Wareham, 2016; Nachira, Nicolai, Dini, Le Louarn, & Leon, 2007; Parker, Van Alstyne, & Jiang, 2016; Tiwana, 2013). We build on recent developments to understand the role IT plays in how business ecosystems *emerge*, reveal important stratification mechanisms (Markus & Loebbecke, 2013) and account for the dynamic interactions within the network of companies in a business ecosystem (Jacobides et al., 2018; Rong et al., 2018). Our empirical study contributes further to the discourse on how technology gives firms leverage to shape the industry and the ecosystem in which they exist (Cusumano & Gawer, 2002). Our framework suggests that collective and boundary resources act as strategic levers for convergence and platform leadership in a business ecosystem. This framework also illustrates technology-enabled practices that shape innovation communities and ecosystem structures (Wang, 2009).

Platforms play an increasingly important role in the public dissemination of creative content and digital cultural artefacts. Traditional broadcasting and networks act as orchestrating platforms (KBS) to determine the interfaces, standards and protocols for constituents to contribute to the creativity and innovation arising from the ecosystem. Other emergent digital platforms (eg, Kakao) aim to promote, consolidate and disseminate distributed creativity, as the conventional hierarchical structures do not adequately address the breadth and importance of interfirm and consumer relationships. Some platforms also provide leadership. They influence initiatives aimed at leading the internal processes of other platforms. Therefore, they can shape the behaviour of organisations that are outside their boundaries (Ghazawneh & Henfridsson, 2013). Some other platforms, on the other hand, focus on open and inclusive initiatives that shape collective action across the ecosystem (Fleming & Waguespack, 2007). In both scenarios, platforms must dialogue closely with customers so that what is created is what consumers want and are willing to pay for. Hence, our study underscores the critical interactions between a multi-sided platform and its internal and external environments, addressing calls for further IS research endeavours on platforms (Agarwal, Gupta, & Kraut, 2008; Tiwana et al., 2010).

For example, our work adds to the existing body of knowledge on the conditions and the process of self-organisation in complex business ecosystems (McKelvey, Tanriverdi, & Yoo, 2015; Mitleton-Kelly, 2003). In this study, we reveal that creative output is enhanced by the colocation of artists and cultural activities in urban and metro areas as well as consumer co-creation, in which consumers participate creatively both in the production of content and the innovation of services. Our study shows that self-organisation in complex business ecosystems may, therefore, be not so much a new socio-economic order as an evolution of the extant economic and cultural order that accounts for consumers' greater access to the ‘means of production’ through IT. Our framework also provides a model of situated creativity, where contingent aspects condition both the individual and social factors of creativity.

5.1.2 | Boundary spanning practice in business ecosystems

Our framework details the fluid boundaries of business ecosystems, adding to IS literature on boundary spanning. Drawing on the processual, situated and emergent nature of BP, we deepened theory on business ecosystems, the conditions under which they operate, the role of IT, their implications and their outcomes. Our empirical investigation adds a boundary spanning-centred discourse on business ecosystems, with a particular focus on how firms effectively mobilise boundary spanning mechanisms (Kortmann, 2012; Levina & Vaast, 2005; McKelvey et al., 2015) such as command and control, establishing partnerships and connecting global ecosystems to help them thrive. We build on prior studies that insist that digital options are imperative for competitive action in a marketplace (eg, Sambamurthy, Bharadwaj, and Grover (2003)). Our framework shows that, beyond the adoption of technology, there is scope for business ecosystems to be shaped through BP.

Our work contributes to the interdependencies perspective in IS literature and the boundary spanning-centred discourse on business ecosystems. Earlier works (Caglio & Ditillo, 2012; Gerdin, 2005; Macintosh & Daft, 1987) focus on the type of tasks and resource interdependencies formed within relationships in particular domains. IS literature that adopts the interdependencies perspective generally posits that technology offers mechanisms to manage different types of work interdependencies found in interfirm relationships (Kumar & Van Dissel, 1996) and online communities (Lindberg et al., 2016). We contribute to this IS literature by illustrating *how* interdependencies emerge in digitally-driven business ecosystems. The emergent K-pop ecosystem has interdependent and coevolving organisations with dynamic and open boundaries. Hence, our research addresses how interdependencies are tied to the development and digital transformation of business ecosystems.

Building on prior work by Kumar and Van Dissel (1996) and Rockart and Short (2012), our study provides empirical evidence that the unique composition of interdependent firms in a business ecosystem forms new interdependencies and IT capabilities in the process of goods or services production. Our framework advances the body of knowledge in IS by demonstrating how an array of interdependencies are enacted and developed in a business ecosystem, and how one type of interdependency relates to another. We reveal new arrangements that manage the interdependencies that ensue within the ecosystem and enable new forms of resource integration and service provision (Barrett et al., 2015). Our study adds a greater understanding of interdependencies as a concept to describe the intensity of interactions within an organisational structure (Staudenmayer, 1997; Thompson, 1967). By considering the implications of heterogeneity and how one type of interdependency relates to another in a business ecosystem, we also address the lack of empirical studies (Gerdin, 2005) on the relationship between IT and interdependencies.

5.1.3 | Creativity processes as commodity

Based on our findings, we develop the relationship between organisational resources and the nature of innovation in ecosystems (Chesbrough & Brunswicker, 2014). Our empirical investigations provide insights on how these various structures, practices and strategies in a digital business ecosystem can enhance creativity and provide value for its members. On the one hand, we reveal how platforms leverage collaborations and synergistic operating processes among other platforms across ecosystems. On the other hand, we demonstrate how technologies in a business ecosystem deliver valuable products or services for consumers in a non-linear value network. This adds to creativity research in IS, which has primarily drawn from reference disciplines in organisational science. We posit there is an increasing importance of cultural and creative goods and services in the modern economy. We add to the IS literature that posits that business ecosystems have blurring or fluid boundaries and multiple components (Wang, 2009), such that innovating digitally requires shared and open practices.

The emergence of many of these multisided platforms means that creativity can then connect back into the innovation system of the ecosystem. Ecosystems encourage pro-competitive and pro-innovative outcomes. The emergence of social networks among content consumer groups enhances the production and innovation for content creators from a 'bottom-up' mechanism. The implication is that creativity is socially situated. In the domain of new digital media, technology effects and the economic impact of socially networked but unaided creative consumers-as-producers are increasingly essential. Our study builds on the earlier work of Baldwin and Von Hippel (2011), who found that customer co-creation manifests in the creative production found in entertainment industries (through video blogging or file sharing) and disrupts the old institutionalised form of creativity and innovations. We found that this old form, which generally maintains clear boundary separation between production and consumption, is now open to creative response and the possibility of new boundary creation.

Through this study, we reveal that both cultural and economic systems evolve and, moreover, that situated creativity is the dynamic intersection of both domains. Similar to earlier works by (Belussi & Sedita, 2008; Potts et al., 2008), we find situated creativity operating over networks, such that downstream markets of mass consumers and producers are connected through creative activity. We reveal that in the case of K-pop music consumer-led social networks are as important as artist-led individual creativity, and that they can lead to creative destruction and structural adjustment that give rise to a new sort of creative industry. While music has always been a social network market, digital technologies arguably allow music to be used in different ways such that it is necessary to rethink arguments about how value is generated. This process is occurring fast in South Korea and where K-pop is popular because the institutional software that allowed the recorded music industry to develop in Western markets – particularly copyright – has been absent as new technologies for copying and sharing have become available.

5.1.4 | Consumption of culture

The global K-pop industry may be instructive when it comes to understanding the processes of creativity, innovation and value generation in a digital context. Our empirical investigation of the industry across several countries in the Asia-Pacific reveals strategies and practices found in rapidly developing business ecosystems in the region (Rong et al., 2018) which have managerial implications for both the East and West. We argue that active processes of innovation, identity formation, creation and communication are integral to acts of cultural consumption across countries.

In this study, we reveal that the digital transformation of business ecosystems gives rise to innovation, policy and reorganisation in response to changes in environmental factors. In this examination of South Korea's K-pop music and entertainment industry, intellectual property was not enforced in South Korea until K-pop emerged. Similar to Potts et al. (2008), we reveal that a lack of alternatives for controlled mass distribution and centralised revenue collection for music copyright owners suggests that other alternatives, such as mobile music and music streaming platforms, have become an important source of revenue for both content providers and creators. Rather than simply re-creating the dynamics of the Western music industry in a mobile world, K-pop, with its new media form, has its own characteristics, and these characteristics influence how music is created and appreciated as well as the demographics of the groups who consume it. The parallels between the world of fashion, which is a pure social network market (Potts et al., 2008), and the emerging world of mobile music may, therefore, be instructive when it comes to understanding processes of creativity, innovation and value generation in the digital context.

In summary, this article provides a starting point for understanding practice-based IT instantiations in the development of self-organising and complex business ecosystems. Drawing on the processual, situated and emergent nature of BP, our work takes up and contributes to theory on the digital transformation of business ecosystems, the conditions under which they operate, the role of IT and their implications and outcomes. Our study contributes to (a) the richness of discourses on boundary spanning in business ecosystems, especially around the topic of how to effectively mobilise boundary spanning mechanisms and IT to enable them to thrive, (b) elucidating the role of orchestrators to develop transient networks that match connectivity and adaptiveness of interconnected industry segments, (c) discovering how new arrangements that manage the contradictions and tensions that ensue within the ecosystem enable new forms of resource integration and service provision and (d) better understanding the relationship between organisational resources and the nature of innovation in ecosystems – practice-based approaches can help scholars better understand IT instantiations.

6 | CONCLUSION

In closing, our study uncovers important insights into the process of digital transformation of business ecosystems. With evidence from an in-depth case study of IT-enabled transformations within the K-pop industry, this study addresses how IT enables effective BP in the transformation of business ecosystems. Our findings also somewhat address the question raised in prior literature of if and how resources can be combined to effectively develop business ecosystems. We reveal how multisided platforms leverage technological capabilities to manage operational interdependencies found across businesses in a business ecosystem. We find that firms in business ecosystems that recognise the importance of bundling IT-enabled resources and mechanisms during product development are more likely to gain specific competencies and capabilities that work to the firm's advantage.

This study is not without its limitations. First, a common criticism of the case study research approach is the problem of transferability or generalizability (Walsham, 2006). While we acknowledge that our findings are drawn from a single context, we contend that our findings on BP for business ecosystems are nevertheless generalisable beyond the current context, and, moreover, that they are corroborated by, and built on, the findings of other studies in the literature. A second limitation of this study concerns the dependability (Trochim, 2006) of our interview data, given the general inability of any researcher to fully account for the dynamic context within which research occurs. Because we were restricted in whom we could speak to, our interview data may be susceptible to bias and errors of recall. Future case studies can reinforce existing findings and reveal perspectives that may have been missed.

Nonetheless, our empirical investigations and data analysis make several research contributions. Notably, our research establishes a platform for a systematic program of research to crystallise business ecosystems and platforms in IS literature. Our findings lead the way for further research into the mechanisms and consequences of IT-enabled boundary spanning, as well as research on structural synergies and antagonisms between the interdependent relationships in business ecosystems.

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ENDNOTES

[1] https://www.nytimes.com/2018/05/28/arts/music/bts-no-1-billboard.html

[2] Other forms of ecosystems include platform ecosystems (Gawer & Cusumano, 2008; Tiwana, Konsynski, & Bush, 2010), innovation ecosystems (Adner & Kapoor, 2010) (Kapoor & Lee, 2013) (Dedehayir, Mäkinen, & Roland Ortt, 2018), entrepreneur ecosystems (Alvedalen & Boschma, 2017) (Hayter, Nelson, Zayed, & O'Connor, 2018) and knowledge ecosystems (Valkokari, 2015).

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[4] http://world.time.com/2012/03/07/south-koreas-greatest-export-how-k-pops-rocking-the-world/

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